ASE, EFM3D & EVL: Datasets, Models & Tools for NBV

Towards Relative Reconstruction Metrics for Next-Best-View

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VCML Seminar WS24/25

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Aria Synthetic Environments

Dataset for Egocentric 3D Scene Understanding

Figure 1: [Ave+24]

Aria Synthetic Environments (ASE)

Dataset Content

- 100,000 unique multi-room interior scenes
- ~2-min egocentric trajectories per scene
- Populated with 8,000 3D objects
- Aria camera & lens characteristics

Ground Truth Annotations

- 6DoF trajectories
- RGB-D frames
- 2D panoptic segmentation
- Semi-dense SLAM PC w/ visibility info
- 3D floor plan (SSL format)
- *GT meshes* as `.ply` files

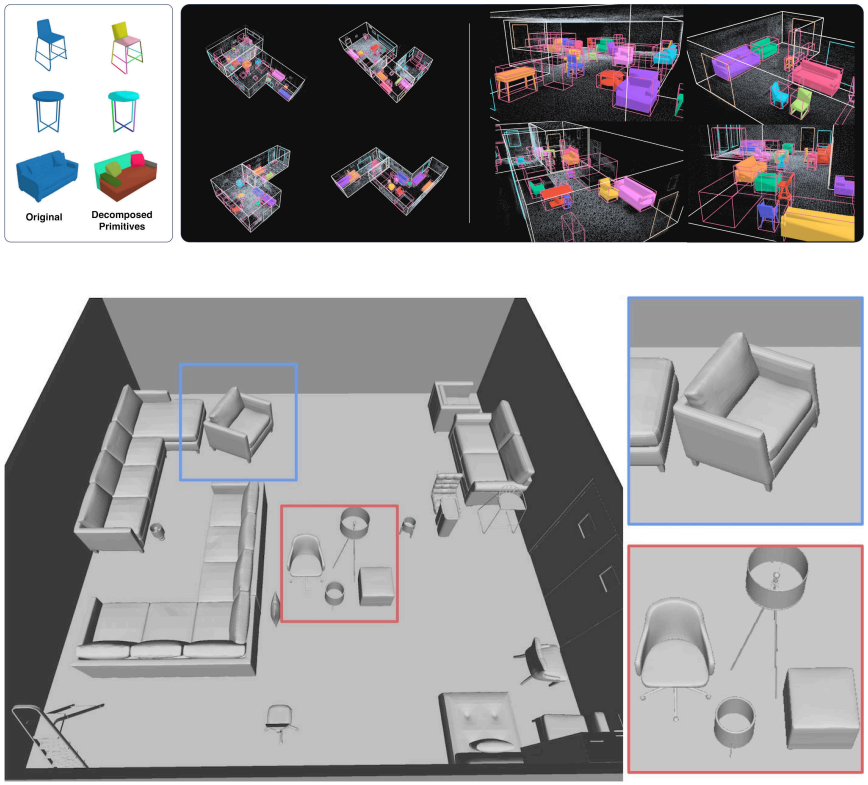

Figure 2: [Ave+24]

Key Resources

- Project Aria Tools for data access
- ASE documentation [Ave+24, Met25a]

ASE Dataset Structure

```
scene_id/
├── ase_scene_language.txt          # Ground truth scene layout in SSL format
├── object_instances_to_classes.json # Mapping from instance IDs to semantic classes
├── trajectory.csv                   # 6DoF camera poses along the egocentric path
├── semidense_points.csv.gz          # Semi-dense 3D point cloud from MPS SLAM
├── semidense_observations.csv.gz    # Point observations (which images see which points)
├── rgb/                             # RGB image frames
│   ├── 000000.png
│   └── ...
├── depth/                           # Ground truth depth maps
│   ├── 000000.png
│   └── ...
└── instances/                       # Instance segmentation masks
    ├── 000000.png
    └── ...
```

EFM3D Benchmark

3D Egocentric Foundation Model:
Egocentric Voxel Lifting (EVL)

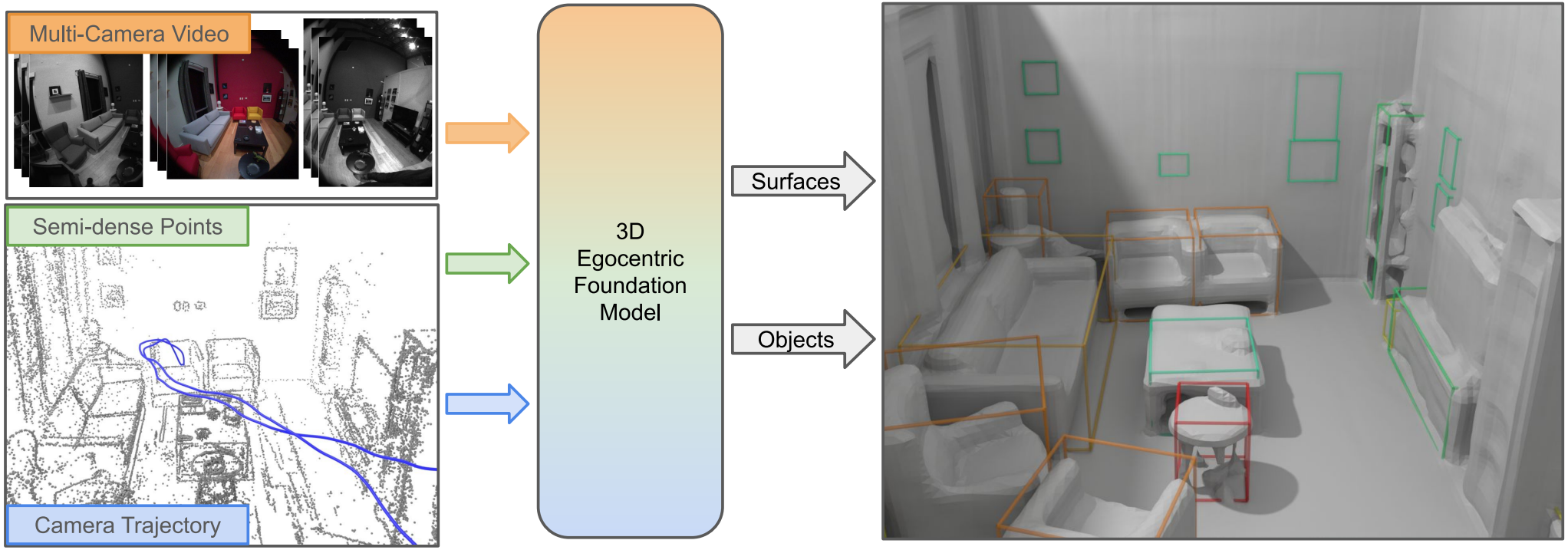

Figure 3: [Str+24]

EFM3D & EVL

EFM3D Tasks

- 3D object detection
- 3D surface regression (occupancy volumes)
  - on ASE, ADT[1], AEO[2] datasets

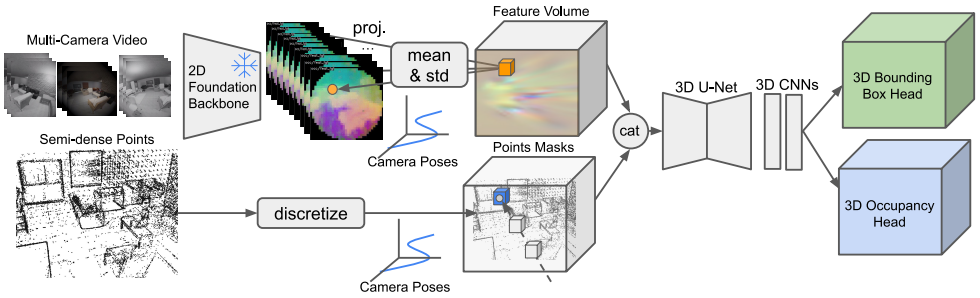

Figure 4: [Str+24]

[1] Aria Digital Twin

[2] Aria Everyday Objects: small-scale, real-world w/ 3D OBBs

EFM3D & EVL

EVL Architecture

- Utilizes **all** available egocentric modalities:
  1. multiple (rectified) RGB, grayscale, and semi-dense points inputs
  2. camera intrinsics and extrinsics
- 16.7M trainable + 86.6M frozen params
- Inherits foundational capabilities from frozen 2D model (DinoV2.5) by lifting 2D features to 3D [Str+24]

EVL: Egocentric Voxel Lifting Architecture

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Model Overview

Egocentric Voxel Lifting (EVL): Multi-task 3D perception from egocentric video
Key Principle: Lift 2D image features to 3D voxel space using camera geometry

EVL: Egocentric Voxel Lifting Architecture

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Input Formulation

$$\boldsymbol{X}_{\text{in}} = \{\boldsymbol{I}_1, \boldsymbol{I}_2, ..., \boldsymbol{I}_F, \boldsymbol{D}_{\text{semi}}, \boldsymbol{K}, \boldsymbol{T}\}$$

Where:

- $\boldsymbol{I}_f \in \mathbb{R}^{H \times W \times 3}$: RGB frames ($F$ frames)
- $\boldsymbol{D}_{\text{semi}} \in \mathbb{R}^{N \times 3}$: Semi-dense 3D points
- $\boldsymbol{K} \in \mathbb{R}^{3 \times 3}$: Camera intrinsics matrix
- $\boldsymbol{T}_f \in \mathrm{SE}(3)$: Camera pose for frame f

Multiple camera streams supported:

- RGB (high-res)
- SLAM cameras (grayscale, rectified)

Output Formulation

3D Occupancy Volume:

$$\boldsymbol{V}_{\text{out}} \in \mathbb{R}^{D_x \times D_y \times D_z \times C}$$

- Voxel grid dimensions: $D_x \times D_y \times D_z$
- C channels for:
  1. Occupancy probability
  2. Object class scores
  3. Surface normals

Detected Objects:

$$\mathcal{O} = \left\{ \left(\boldsymbol{b}_i^{\text{3D}}, c_i, s_i \right) \right\}_{i=1}^{N}$$

- $\boldsymbol{b}_i^{\text{3D}} \in \mathbb{R}^9$: Oriented bounding box
- c_i: Object class
- s_i: Confidence score

EVL: Egocentric Voxel Lifting Architecture

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Feature Lifting Process

1. **2D Feature Extraction**: Frozen DinoV2.5 backbone

$$\boldsymbol{F}_{\text{2D}} = \varphi_{\text{DINOv2.5}(\boldsymbol{I}_f)} \in \mathbb{R}^{H' \times W' \times D_{\text{feat}}}$$

2. **3D Projection**: For each voxel $\boldsymbol{v} \in \mathbb{R}^3$, aggregate features from all frames

$$\boldsymbol{F}_{\text{3D}(\boldsymbol{v})} = \text{Aggregate}\left(\left\{\pi\left(\boldsymbol{T}_f^{-1}\boldsymbol{v}, \boldsymbol{K}, \boldsymbol{F}_{\text{2D}}^f\right)\right\}_{f=1}^{F}\right)$$

   where $\pi(\cdot)$ is the camera projection function

3. **3D Convolution**: Process lifted features

$$\boldsymbol{V}_{\text{out}} = \psi_{\text{3D-CNN}(\boldsymbol{F}_{\text{3D}}, \boldsymbol{D}_{\text{semi}})}$$

ATEK Toolkit

Streamlined ML Workflows for Aria Datasets

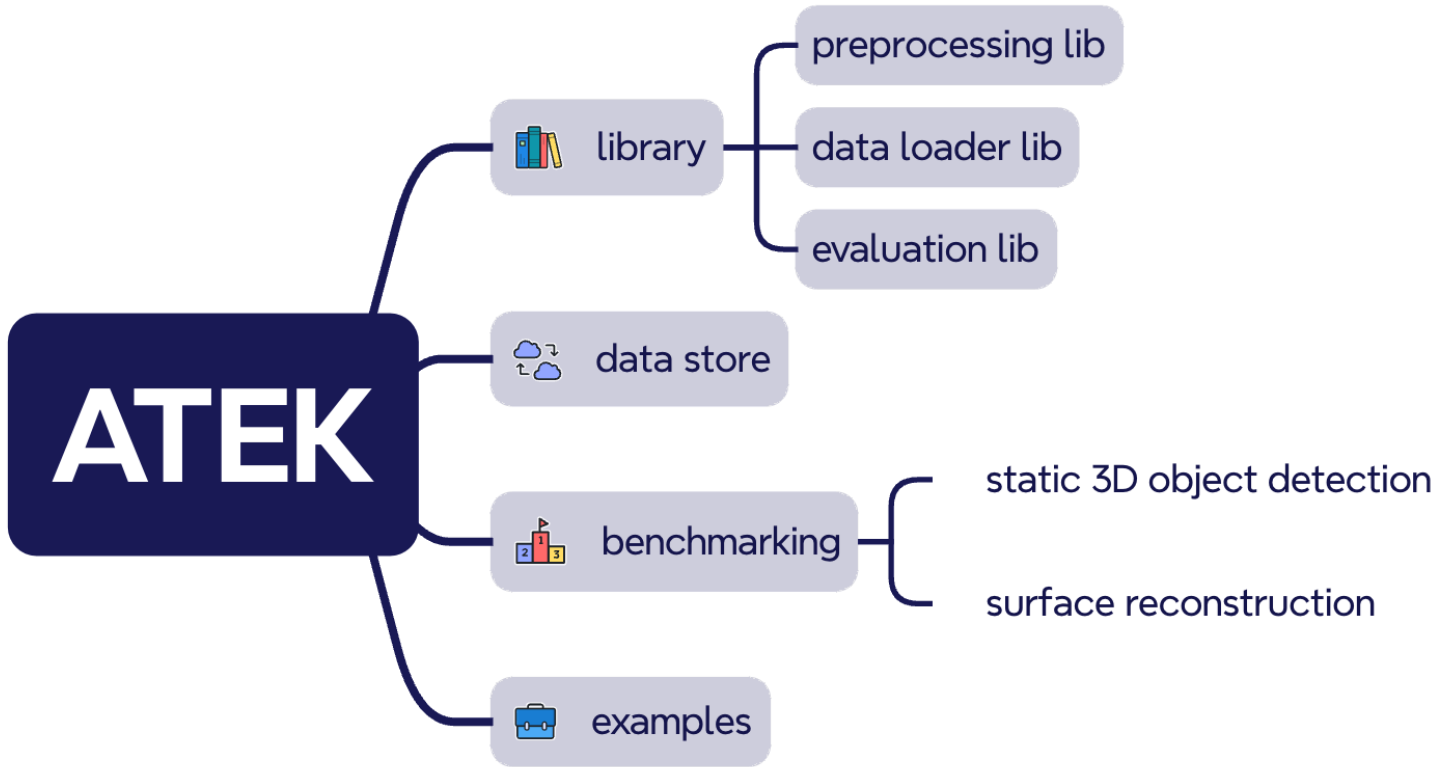

Figure 5: [Met25b]

ATEK Toolkit

ATEK Data Store

- Pre-processed for various tasks → ready for PyTorch training
- Local download or cloud streaming
- Eval metrics (accuracy, completeness, F-score) → adaptation for RRI
- Integration w/ Meta's MPS
- Various example notebooks

Provided Models

- *Cube R-CNN* [Bra+23] for OBBs
- *EFM* [Str+24] for OBBs & surface reconstruction

Resources

- ATEK GitHub [Met25c]
- ECCV 2024 Tutorial: Egocentric Research with Project Aria
- Atek Context7 ID: `/facebookresearch/atek`

ATEK provides streamlined ML workflows for rapid prototyping and benchmarking on Aria datasets.

Next Steps & TODOs

Literature Review

- Read Project Aria paper [Met25a]
- Study EFM3D & EVL architecture in depth [Str+24]
- Deep dive into GenNBV's multi-source embeddings [Che+24]
- Compare VIN-NBV vs. GenNBV: RRI prediction vs. coverage-based rewards

Technical Exploration

- Explore GT meshes (`.ply` files) in ASE dataset
- Get familiar with ATEK and ATEK Data Store
- Test mesh-based evaluation metrics (accuracy, completeness, F-score)
- Experiment with probabilistic 3D occupancy grids

Next Steps & TODOs

Implementation Goals

- Implement ray-casting for mesh-based visibility computation
- Develop entity-wise RRI computation pipeline using GT meshes
- Design 5DoF action space for scene exploration
- Build multi-source state embedding (geometric + semantic + action)
- Prototype RRI prediction network architecture

Reconstruction Quality Metrics Theory

Surface-to-Surface Distance Metrics

Accuracy (Prediction → GT):

$$\mathrm{Acc} = \frac{1}{|\mathcal{P}|} \sum_{\boldsymbol{p} \in \mathcal{P}} \min_{\boldsymbol{q} \in \mathcal{M}_{\mathrm{GT}}} \|\boldsymbol{p} - \boldsymbol{q}\|_2$$

Completeness (GT → Prediction):

$$\mathrm{Comp} = \frac{1}{|\mathcal{M}_{\mathrm{GT}}|} \sum_{\boldsymbol{q} \in \mathcal{M}_{\mathrm{GT}}} \min_{\boldsymbol{p} \in \mathcal{P}} \|\boldsymbol{p} - \boldsymbol{q}\|_2$$

Where:

- $\mathcal{P}$: Sampled points from predicted mesh
- $\mathcal{M}_{\mathrm{GT}}$: Sampled points from ground truth mesh
- Typically sample 10k points from each mesh

Precision, Recall & F-score

At threshold τ (typically 5cm):

$$\mathrm{Precision@}\tau = \frac{\left|\left\{\boldsymbol{p} \in \mathcal{P} : \min_{\boldsymbol{q} \in \mathcal{M}_{\mathrm{GT}}} \|\boldsymbol{p} - \boldsymbol{q}\| < \tau\right\}\right|}{|\mathcal{P}|}$$

$$\mathrm{Recall@}\tau = \frac{\left|\left\{\boldsymbol{q} \in \mathcal{M}_{\mathrm{GT}} : \min_{\boldsymbol{p} \in \mathcal{P}} \|\boldsymbol{p} - \boldsymbol{q}\| < \tau\right\}\right|}{|\mathcal{M}_{\mathrm{GT}}|}$$

$$\text{F-score@}\tau = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

Chamfer Distance (Bidirectional)

$$\mathrm{CD}(\mathcal{P}, \mathcal{M}_{\mathrm{GT}}) = \mathrm{Acc} + \mathrm{Comp}$$

Combines both directions of surface error

Reconstruction Quality Metrics Theory

ATEK Implementation: `evaluate_single_mesh_pair()` computes all metrics using trimesh sampling and PyTorch3D distance functions [Met25c]

Computing RRI with GT Meshes

RRI Formulation for Mesh-based Reconstruction

Given:

- $\mathcal{M}_{\text{GT}}$: Ground truth mesh (from ASE `.ply` files)
- $\mathcal{M}_t$: Current partial reconstruction at step t
- v_{cand}: Candidate viewpoint
- $\mathcal{M}_{t \cup v}$: Updated reconstruction after capturing from v_{cand}

Mesh-based RRI:

$$\text{RRI}(v_{\text{cand}}) = \frac{d(\mathcal{M}_t, \mathcal{M}_{\text{GT}}) - d(\mathcal{M}_{t \cup v}, \mathcal{M}_{\text{GT}})}{d(\mathcal{M}_t, \mathcal{M}_{\text{GT}})}$$

where $d(\cdot, \cdot)$ can be:

- Accuracy (one-sided distance)
- Completeness (reverse distance)
- Chamfer Distance (bidirectional)
- F-score improvement

Computing RRI with GT Meshes

Entity-wise RRI Aggregation

For multi-entity scenes:

$$\mathrm{RRI}_{\mathrm{total}(\boldsymbol{v})} = \sum_{e \in \mathcal{E}} w_e \cdot \mathrm{RRI}_e(\boldsymbol{v})$$

Where:

- $\mathcal{E}$ = {walls, doors, windows, objects, ...}
- w_e: Semantic importance weight
- RRI_e: RRI computed per entity using entity-specific GT mesh

Visibility-based RRI Oracle

Using ASE visibility data:

1. Ray-cast from $\boldsymbol{v}_{\mathrm{cand}}$ to $\mathcal{M}_{\mathrm{GT}}$
2. Identify newly visible mesh faces
3. Estimate improvement without actual capture:

$$\mathrm{RRI}_{\mathrm{oracle}(\boldsymbol{v})} \approx \frac{A_{\text{visible-new}}}{A_{\text{total-unseen}}}$$

where A denotes surface area

VIN-NBV: Learning-Based Next-Best-View

Key Innovation

- First NBV method to directly optimize **reconstruction quality** (not coverage)
- Predicts Relative Reconstruction Improvement (RRI) without capturing new images
- 30% improvement over coverage-based baselines
- Trained 24h on 4 A6000 GPUs [Fra+25]

VIN Architecture

Predicts RRI from current reconstruction state:

$$\widehat{\mathrm{RRI}}(\boldsymbol{q}) = \mathrm{VIN}_\theta(\mathcal{R}_{\mathrm{base}}, \boldsymbol{C}_{\mathrm{base}}, \boldsymbol{C}_{\boldsymbol{q}})$$

- **Input**: Partial point cloud + camera poses
- **Features**: Surface normals, visibility counts, depth, coverage
- **Output**: Predicted RRI via ordinal classification (15 bins)

VIN-NBV: Learning-Based Next-Best-View

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Relative Reconstruction Improvement

For a candidate view $\boldsymbol{q}$, RRI quantifies expected improvement:

$$\mathrm{RRI}(\boldsymbol{q}) = \frac{\mathrm{CD}(\mathcal{R}_{\mathrm{base}}, \mathcal{R}_{\mathrm{GT}}) - \mathrm{CD}\big(\mathcal{R}_{\mathrm{base}} \cup \boldsymbol{q}, \mathcal{R}_{\mathrm{GT}}\big)}{\mathrm{CD}(\mathcal{R}_{\mathrm{base}}, \mathcal{R}_{\mathrm{GT}})}$$

- Range: $[0, 1]$ where higher = better view
- Normalized by current error $\rightarrow$ scale-independent
- CD measures reconstruction quality

VIN-NBV demonstrates that *learning reconstruction-aware NBV policies* significantly outperforms traditional coverage-based approaches.

GenNBV: Generalizable Next-Best-View Policy

Key Innovations

- **5DoF free-space action space**: 3D position + 2D rotation (yaw, pitch)
- Multi-source state embedding: geometric, semantic, action representations
- *Probabilistic 3D occupancy grid* vs. binary (distinguishes unscanned from empty)
- Cross-dataset generalization: 98.26% coverage on Houses3K, 97.12% on OmniObject3D

State Representation

Geometric Embedding s_t^G:

- Probabilistic 3D occupancy grid from depth maps
- Bresenham ray-casting with log-odds update
- Three states: occupied, free, unknown

Semantic Embedding s_t^S:

- RGB images $\longrightarrow$ grayscale $\longrightarrow$ 2D CNN
- Helps distinguish holes from incomplete scans

Action Embedding s_t^A:

- Historical viewpoint sequence encoding

Combined: $s_t = \mathrm{Linear}\left(s_t^G; s_t^S; s_t^A\right)$

GenNBV: Generalizable Next-Best-View Policy

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Action Space Design

$$\mathcal{A} = \underbrace{\mathbb{R}^3}_{\text{position}} \times \underbrace{SO(2)}_{\text{heading}}$$

- Approximately 20m × 20m × 10m position space
- Omnidirectional heading subspace
- *No hand-crafted constraints* (e.g., hemisphere)

RL-based framework with PPO. Reward: Δ CR between steps. [Che+24]

Synthesis: RRI-based NBV for Scene-Level Reconstruction

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Combining VIN-NBV & GenNBV Insights

From VIN-NBV: Direct reconstruction quality optimization via RRI
From GenNBV: Free-space exploration + multi-source state embedding
Our Approach: Adapt RRI prediction to scene-level environments with 5DoF action space

Synthesis: RRI-based NBV for Scene-Level Reconstruction

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RRI with GT Meshes

Use ASE visibility data + GT meshes for *oracle RRI*:

$$\mathrm{RRI}(\boldsymbol{q}) = \frac{d(\mathcal{M}_{\text{partial}}, \mathcal{M}_{\text{GT}}) - d(\mathcal{M}_{\text{partial}} \cup \boldsymbol{q}, \mathcal{M}_{\text{GT}})}{d(\mathcal{M}_{\text{partial}}, \mathcal{M}_{\text{GT}})}$$

where $\mathcal{M}$ represents meshes, $d(\cdot, \cdot)$ is mesh distance

Entity-wise RRI:

$$\mathrm{RRI}_{\text{total}} = \sum_{e \in \mathcal{E}} w_e \cdot \mathrm{RRI}_e$$

where $\mathcal{E}$ = {walls, doors, objects, ...}

Proposed Pipeline

1. **Reconstruct**: Build partial mesh from historical trajectory
2. **Sample**: Generate candidate viewpoints in free space
3. **Compute Features**: Extract geometric + semantic embeddings (à la GenNBV)
4. **Predict**: Use network to predict RRI per candidate
5. **Select**: Choose NBV with highest predicted RRI

> **Key Challenge**: Ray-casting from candidate views to compute visibility on GT meshes for entity-wise RRI computation

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